

Drug-Target Interaction Prediction Using Multi-Head Self-Attention and Graph Attention Network

Zhongjian Cheng, Cheng Yan^{ID}, Fang-Xiang Wu^{ID}, and Jianxin Wang^{ID}

Abstract—Identifying drug-target interactions (DTIs) is an important step in the process of new drug discovery and drug repositioning. Accurate predictions for DTIs can improve the efficiency in the drug discovery and development. Although rapid advances in deep learning technologies have generated various computational methods, it is still appealing to further investigate how to design efficient networks for predicting DTIs. In this study, we propose an end-to-end deep learning method (called MHSADTI) to predict DTIs based on the graph attention network and multi-head self-attention mechanism. First, the characteristics of drugs and proteins are extracted by the graph attention network and multi-head self-attention mechanism, respectively. Then, the attention scores are used to consider which amino acid subsequence in a protein is more important for the drug to predict its interactions. Finally, we predict DTIs by a fully connected layer after obtaining the feature vectors of drugs and proteins. MHSADTI takes advantage of self-attention mechanism for obtaining long-dependent contextual relationship in amino acid sequences and predicting DTI interpretability. More effective molecular characteristics are also obtained by the attention mechanism in graph attention networks. Multiple cross validation experiments are adopted to assess the performance of our MHSADTI. The experiments on four datasets, human, *C.elegans*, DUD-E and DrugBank show our method outperforms the state-of-the-art methods in terms of AUC, Precision, Recall, AUPR and F1-score. In addition, the case studies further demonstrate that our method can provide effective visualizations to interpret the prediction results from biological insights.

Index Terms—Drug-target interactions, multi-head self-attention, graph attention network

1 INTRODUCTION

THE accurate identification of drug-target interactions (DTIs) is a critical task for new drug discovery and drug repositioning. Because biological experiments cause excessive time and expensive lab cost [1], predicting potential DTIs through efficient computational methods, which can reduce the scope of conducting biological experiments, is demanded urgently. In recent years, although many effective prediction methods have been proposed by researchers, how to design more accurate and efficient methods for predicting DTIs is still a major challenge.

In order to accurately predict potential DTIs, many machine learning-based methods have been proposed [2], [3], [4], [5], [6], [7], [8], [9]. These methods can use

more data from both drugs and targets in a unified framework simultaneously from chemogenomic perspective [10]. For example, Bleakley and Yamanishi [2] develop a bipartite local model which applies SVMs on known interactions after employing the similarity measurements between protein sequences and chemical structures. Cheng *et al.* [3] employ the multi-target quantitative structure-activity relationships (mt-QSAR) and computational chemogenomics for predicting DTIs based on feature selection techniques. Van Laarhoven and Marchiori [4] propose a novel model which integrates a simple weighted nearest neighbor method into a machine learning method previously proposed and get a high accuracy prediction result. Wang and Zeng [5] propose a model based on restricted Boltzmann machines. Compared with the method that only mixes multiple types of interactions, their model achieves the better prediction performances for different types of DTIs. Yuan *et al.* [6] propose a new method (called DrugE-Rank) to improve the prediction performance and a variety of well-known similarity-based methods are used as components of ensemble learning in DrugE-Rank. Based on the heterogeneous map of known DTIs with multiple similarities between drugs and multiple similarities between targets, Olayan *et al.* [7] develop a novel method (called DDR) which applies a non-linear similarity fusion method to multiple similarities selected in a heuristic process, then they extract the features from the heterogeneous graph and train a random forest to predict DTIs.

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Lee and Nam [8] use a random walk with restart algorithm to utilize global network topology information for solving the problem which only considers directly connected nodes on the concept of “guilt-by-association”. Mohamed *et al.* [9] regard the problem as a link prediction in knowledge graphs and propose a knowledge graph embedding model (called TriModel) to learn vector representations for downstream. Because the traditional machine learning-based methods are relatively simple, they may not be able to learn the complicated concepts in the datasets which contain a large amount of data and the methods based on machine learning gradually perform poorly on large amounts of data.

Recently, deep learning-based methods have achieved excellent results for lots of prediction problems. Many end-to-end deep learning methods have also been used for predicting DTIs and achieved better performance than traditional machine learning methods [11], [12], [13], [14], [15], [16]. Compared to traditional feature selection methods, Wang *et al.* [11] use the stacked autoencoder to mine the information from raw data adequately and the method generates highly representative features automatically. By representing proteins with 2D distance maps from the monomer structure, Zheng *et al.* [16] follow the Visual Question Answering mode and propose a deep learning framework which extracts information of drugs by the Bi-LSTM structure and characteristics of proteins by dynamic CNN structure. Lin *et al.* [12] introduce a probabilistic model named FNML for dealing with overwhelming negative samples and it achieves a robust performance on highly imbalanced datasets of DTIs. Tsubaki *et al.* [13] propose an end-to-end representation learning model for DTIs which extracts representative features of drugs and proteins by a graph neural network and a convolution neural network (CNN), respectively. Although the method achieves very good performance, it is obvious that the amino acid sequences of proteins have association information from the context in real biological environments. There are many potential interactions between different positions in the sequences. However, the convolutional window of CNN is fixed and then CNN structure can not model these characteristics of contextual information. Based on a two-way attention mechanism, Gao *et al.* [14] propose an interpretable model which uses the long short-term memory (LSTM) to extract contextual information for amino acid sequences and graph convolutional networks (GCN) to obtain feature vectors of drugs. The LSTM structure effectively solves the problem that the contextual information of amino acid sequences can not be obtained. However, as the length of amino acid sequences are usually large, LSTM can not learn long-dependent contextual association information. Zheng *et al.* [15] propose a hybrid model that integrates the CNN and LSTM for DTIs, similarly, it also does not solve these problems.

In this study, we propose a model (called MHSADTI) to predict DTIs with the multi-head self-attention mechanism [17] and graph attention networks (GATs) [18]. When extracting the characteristics of amino acid sequences of proteins, the multi-head self-attention mechanism can solve the problem that CNN can not obtain contextual association information. At the same time, it can also gain the long-

TABLE 1
Summary of the Four Datasets, Human, *C.elegans*, DUD-E and DrugBank

Datasets	Number of Proteins	Number of Drugs	Number of total sample	Number of positive interactions
human	852	1,052	6,738	3,369
<i>C.elegans</i>	2,504	1,434	8,000	4,000
DUD-E	20,489	102	68,070	22,886
DrugBank	4,794	6,707	37,283	18,816

dependent information in the sequences which can not be learned by RNN, LSTM and other structures when the lengths of amino acid sequences are too long. When extracting the characteristics of drugs, we use the GAT instead of the graph convolutional network. This is advantage that GAT can specify implicitly different weights to different nodes in a neighborhood which can avoid the possible effects of noise connection on nodes in the graph. The GAT model further enhances the validity of the representative feature vectors of the graphs. Finally, we also use attention scores to help us consider which amino acid subsequence in a protein is more important for the drug to predict its interaction through drug-protein attention network modules [13]. Evaluated on the four datasets, human, *C.elegans*, DUD-E and DrugBank, the proposed model achieves the better performances than state-of-the-art methods in terms of AUC (area under the receiver operating characteristics curve), Precision, Recall, AUPR (area under the precision-recall curve) and F-score. The results illustrate the validity of the proposed model. In order to help understand what the network structure has been learned intuitively, we provide the visualized results for multi-head self-attention and drug-protein attention.

2 DATASETS

In the task of predicting DTIs, randomly selected negative samples may include potentially positive samples, which causes the model to have higher accuracy during the training phase but not perform well in the test datasets [19]. So an effective sampling method for negative samples in datasets is critical to predict DTIs. However, there are many existing datasets with only random negative sampling have been applied to evaluate the performance of algorithms for predicting DTIs [20], [21].

In this study, we focus on evaluating the model on two different datasets (human and *C.elegans*). The two datasets are created by an effective negative sampling method [22], where human dataset contains 3,369 positive interactions between 1,052 unique compounds and 852 unique proteins; the *C.elegans* dataset contains 4,000 positive interactions between 1,434 unique compounds and 2,504 unique proteins. Based on the assumption that similar compounds interact with the proteins that are similar to known proteins, more credible negative samples are obtained after applying specific dissimilarity rules to the screening framework. Table 1 shows the details of two datasets. The human dataset has been used in the DeepCPI [13], [16] and [22]. The *C.elegans* has been used in the [13] for predicting DTIs. In the

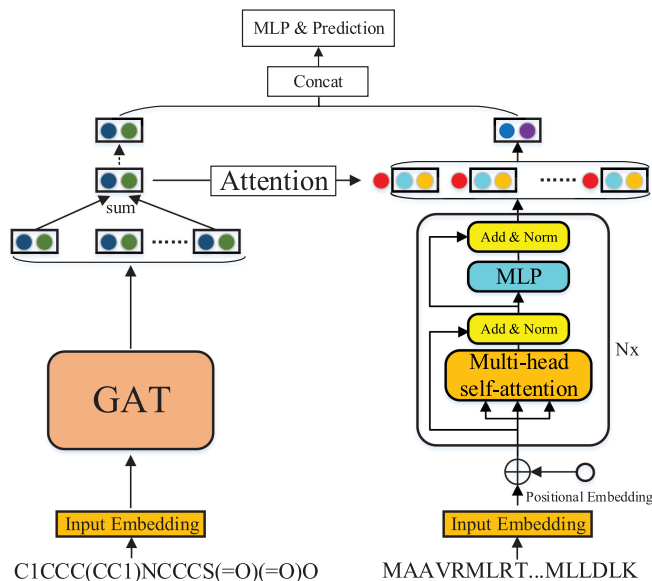


Fig. 1. An overview of the proposed prediction approach.

experiments, we get the two datasets created from [13] whose ratio of positive and negative samples is 1:1. The two datasets can be obtained from the link address (https://github.com/masashitsubaki/CPI_prediction).

In order to further illustrate the accuracy of the proposed model, we also conduct the experiment on the DUD-E benchmark, a robust dataset for structure-based virtual screening methods. It contains 102 diverse target proteins (provided as PDB files), 22,886 active pairs of proteins and compounds (provided as SMILES (simplified molecular input line entry specification)). The DUD-E dataset has been used in [16] and [23]. We follow the experimental setting in [23] and the dataset is available at (<https://github.com/prokia/drugVQA>). Since the training, validation and test sets of DUD-E are well-divided and the competing method (DeepCPI [13]) has also been evaluated with three-fold cross validation in previous works, we also conduct three-fold cross validation to fairly evaluate and compare the performances of methods on DUD-E.

Compared to smaller datasets (human and *C.elegans*), we also test our proposed method on datasets with larger amounts of data. We organize the data in the DrugBank database and obtain a dataset containing a large amount of data. The DrugBank dataset contains 18,816 positive interactions between 4,794 unique compounds and 6,707 unique proteins. Finally, the prediction performances of our method and competing methods are evaluated in terms of AUC, Precision, Recall, AUPR and F1-score on the above four datasets.

3 METHOD

In this study, we propose a novel approach, named MHSADTI, to predict DTIs. We first give the brief description of the approach, and then focus on the different modules of the approach.

Fig. 1 shows an overview of the proposed prediction approach. The feature vectors of drug molecules and proteins, which are low-dimensional real-valued representations obtained using a GAT (Section 3.1) and an encoder

based on multi-head self-attention mechanism (Section 3.2), are concatenated as input to the classifier for predicting DTIs. The input of the algorithm is the raw data of drug molecules and proteins (SMILES and amino acid sequences). After the embedding layers, molecular graphs obtained from SMILES preprocessed by RDKit are input to GAT and protein sequences based on n -gram are input to the encoder.

3.1 Graph Attention Network for Drugs

With the advent of graph structure data in recent years, many methods have focused on the representation and learning of graphs, and thus great progress has been made in graph-based studies [24], [25], [26], [27]. Compared to grid-like structure data, graph structure data usually lies in an irregular domain which makes it more difficult to accurately represent the characteristics of graphs. When modeling the characteristics of drugs, drug molecules represented by SMILES can be transformed into the graphs of molecules. In the graph, the atoms represent nodes and chemical bonds between atoms are represented as existing edges. Finally, we can effectively extract map information based on existing works [24], [25], [26], [27] which can help us complete prediction tasks.

Due to the difficulties of representation learning in molecules which have few types of atoms and chemical bonds, we use the r -radius subgraphs [28] to get the vertical integer encoding sequence of each drug. These subgraphs contain large amounts of information about the drug molecules for fully learning parameters in networks. Using the same procedure as [13], we can get a set of vertex features after embedding for each drug molecule, $D = \{d_1, d_2, \dots, d_{|D|}\}$, $d_i \in R^d$, where d is the embedding dimension. Through the transformation of a weight matrix, $W \in R^{F \times d}$, we can obtain more representation information in the high dimension than the original vector space. In order to learn the structural information of the graph, a self-attention mechanism is applied to the graph and the attention coefficient can be obtained as follows.

$$e_{ij} = a(Wd_i, Wd_j) \quad j \in \mathcal{N}_i. \quad (1)$$

The coefficient e_{ij} indicates the importance of vertex j to vertex i in the graph, where $\mathcal{N}_i$ is the neighborhood of vertex i in the graph. Instead of allowing each vertex to attend others, we keep the structural information between atomic vertices by $\mathcal{N}_i$. Next, we normalize them using a *softmax* function

$$a_{ij} = \text{softmax}(e_{ij}) = \frac{\exp(e_{ij})}{\sum_{k \in \mathcal{N}_i} \exp(e_{ik})}. \quad (2)$$

Finally, combining Equations (1) and (2), the attention coefficients are expressed as

$$a_{ij} = \frac{\exp(\text{LeakyRelu}(\alpha^T [Wd_i || Wd_j]))}{\sum_{k \in \mathcal{N}_i} \exp(\text{LeakyRelu}(\alpha^T [Wd_i || Wd_k]))}, \quad (3)$$

where $\alpha \in R^{2 \cdot F}$ is the weight vector, the *softmax* function applies LeakyRelu [29] nonlinear transformation and $||$ indicates the concatenated operation. As a result, we can get the

final output features of each vertex after combining the characteristic information of its neighbors.

$$d'_i = \sigma \left(\sum_{j \in \mathcal{N}_i} a_{ij} W d_j \right), \quad (4)$$

where σ is the *Relu* activation function, W is the dimensional transformation matrix, d_j is the j th neighbor feature vector. Each vertex integrates the neighbors' features according to the attention weights with its neighbors, that is to say, the vertex is represented as a weighted average of its neighbor feature vectors. After the activation function, the semantic feature vector of the vertex is obtained. In order to make the learning process of the self-attention mechanism more stable and effective, we introduce the idea of multi-attention, averaging the feature vectors learned by multiple attentions to obtain the final feature vectors of the hidden layer.

$$d''_i = \sigma \left(\frac{1}{H} \sum_{h=1}^H \sum_{j \in \mathcal{N}_i} a_{ij}^h W^h d_j \right), \quad (5)$$

where H is the number of attention. By adjusting the dimensions of the transformation matrix W each time, we can ensure that the final transformed output feature dimension is F . After the convolution of t layers, the final feature representation of each vertex in the graph can be obtained, $D^{(t)} = \{d_1^{(t)}, d_2^{(t)}, \dots, d_{|D|}^{(t)}\}$. Averaging the vectors of vertices, we can obtain the feature vector of the drug, y_{drug} as follows.

$$y_{drug} = \frac{1}{|D|} \sum_{i=1}^{|D|} d_i^{(t)}. \quad (6)$$

3.2 Multi-Head Self-Attention for Protein Sequences

The self-attention mechanism has been widely used in deep learning tasks with contextual relationships including machine translation, semantic understanding [30], [31]. Through the parameters which can be learned by the backward propagation in the network, the model is able to relate different positions of the sequence to calculate the importance of the sequence representation. In particular, Transformer [17], the use of self-attention to calculate the representation of its inputs and outputs instead of sequence-aligned RNN or CNN, achieves a major breakthrough in the machine translation. Since the beginning of this work, there are many excellent and advanced research works evolved in the field of natural language processing [32].

In this section, we use the fully connected self-attention module to learn the characteristic information of the amino acid sequences with the encoder structure of Transformer. Along with the effectiveness and robustness of the multi-head self-attention mechanism, this structure addresses some critical problems that CNN can not obtain contextual information in the sequences. That LSTM can not obtain long-dependent information when the sequence is long has also been overcome. In the experimental part, the visualization results learned by the attention mechanism are shown to illustrate the effectiveness of MHSADTI.

3.2.1 Preprocessing and Position Encoding

As explained in [33], we use the n -gram to define the "word" in amino acid sequences. Given an amino acid sequence, by splitting the protein sequences into overlapping n -gram amino acids, we can get its vector representation $P = \{p_1, p_2, \dots, p_{|P|}\}$, where $p_i \in R^d$ is the embedding vector of the i th word and d is the embedding dimension. In the input part of the data, the model containing no recurrence and no convolution can not make full use of the order of the sequence. In order to solve this problem, a "position encoding" is set to obtain relative position information for each "word" in the sequence. The "position encoding" can provide context information for each position of the sequence.

$$PE_{(pos, 2j)} = \sin(pos/10000^{2j/d}) \quad (7)$$

$$PE_{(pos, 2j+1)} = \cos(pos/10000^{2j/d}), \quad (8)$$

where pos and $j \in [0, \frac{d}{2}]$ represent the position and the dimension, respectively. In Equations (7) and (8), the function $PE_{(\bullet)}$ can effectively learn information between relative positions because there is a linear relationship between PE_{pos} and PE_{pos+k} for the fixed offset k .

$$\begin{cases} \sin(\alpha + \beta) = \sin(\alpha) \cos(\beta) + \cos(\alpha) \sin(\beta) \\ \cos(\alpha + \beta) = \cos(\alpha) \cos(\beta) - \sin(\alpha) \sin(\beta) \end{cases} \quad (9)$$

So we can get the linear relationship between the pos position and the $pos+k$ position.

$$PE_{(pos+k, 2j)} = PE_{(pos, 2j)} \times PE_{(k, 2j+1)} + PE_{(pos, 2j+1)} \times PE_{(k, 2j)} \quad (10)$$

$$PE_{(pos+k, 2j+1)} = PE_{(pos, 2j+1)} \times PE_{(k, 2j+1)} - PE_{(pos, 2j)} \times PE_{(k, 2j)}. \quad (11)$$

For each position in the sequence, through the function $PE_{(\bullet)}$, we can model the contextual feature vectors of different positions. Finally, the sum of the "word" embedding vector and the position encoding is used as the input to the encoder for downstream tasks.

3.2.2 Encoder Based on the Multi-Head Self-Attention Mechanism

Fig. 2 shows the detailed process of the self-attention structure. The top and bottom of Fig. 2 are the output vectors processed by the self-attention structure and input vectors preprocessed in the Section 3.2.1, respectively. The part enclosed by the red dotted line is the self-attention learning process between each "word" vector in an amino acid sequence and the "word" vectors in its contextual sequence. As depicted in Fig. 2, each "word" vector can be represented as a query, key-value pair through three mapping matrices whose parameters can be learned by backward propagation. For the output of each position, the attention mechanism maps a query and a set of key-value pairs to get

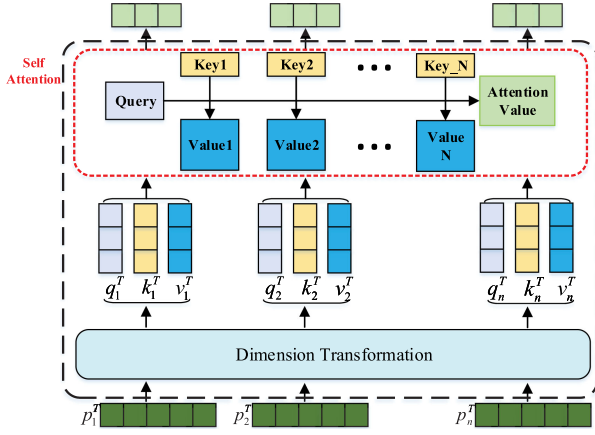


Fig. 2. The detailed process of the self-attention structure.

the weighted sum at different locations in the context. The weight assigned to each value is calculated according to the query and the corresponding key.

More specifically, after preprocessing in Section 3.2.1 and mapping, we can get the representation of query, key, value for each amino acid sequence, $Q = \{q_1^T, q_2^T, \dots, q_L^T\}$, $K = \{k_1^T, k_2^T, \dots, k_L^T\}$, $V = \{v_1^T, v_2^T, \dots, v_L^T\}$, where L represents the length of the sequence and $q_i^T \in R^d$, $k_i^T \in R^d$, $v_i^T \in R^d$. Finally, we can get the output of the i th position after self-attention processing.

$$Attention(q_i, K, V) = softmax\left(\frac{q_i K^T}{\sqrt{d}}\right)V, \quad (12)$$

where $1/\sqrt{d}$ is the scaling factor which is defined to be effective for dealing with the small gradients when d is too large after the *softmax* function [17].

Compared to recurrent neural networks, the model can guarantee the parallel computing of data while learning the contextual information from the sequences and greatly reduce the time complexity for processing large amounts of data. We can calculate the output of the entire self-attention layer simultaneously.

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d}}\right)V. \quad (13)$$

In order to maintain its validity and stability, the model also incorporates the learning results of the multi-attention which is able to learn the various semantic information of the features in different spaces using multiple self-attention weight matrices. Finally, the results learned by the multi-attention mechanism are concatenated as output.

$$head_i = Attention\left(QW_i^Q, KW_i^K, VW_i^V\right) \quad (14)$$

$$MultiHead(Q, K, V) = Concat(head_1, \dots, head_h)W^O, \quad (15)$$

where $W_i^Q \in R^{d \times d_k}$, $W_i^K \in R^{d \times d_k}$, $W_i^V \in R^{d \times d_v}$ and $W^O \in R^{h \times d_v \times d}$ are used for feature space transformation, $Concat(\bullet)$ is the vector concatenated operation.

The second part of the encoder is a fully connected forward network which contains two linear transformation structures applied to each position in the sequence separately and identically.

$$MLP(x) = W_2(relu(W_1x + b_1)) + b_2, \quad (16)$$

where $W_1 \in R^{d_{inner} \times d}$, $W_2 \in R^{d \times d_{inner}}$ are transformation parameters, *relu* is the activation function, x is the output from the first part of encoder.

As shown in Fig. 1, similar to most network structures, in addition to two parts of the encoder, the residual connection [34] structure and the layer normalization structure [35] are used as the input and output. The normalization operations and jump connections can improve the generalization ability of the model. The layer normalization can make the network converging faster by keeping the input of each layer with the same data distribution during the training. The residual connection can get rich characteristic information of hidden layers by retaining not only the information of the last layer but also the output information of the multi-layer network processing. Finally, we can obtain the characteristics of each amino acid sequence after learning the k -layer encoder structure, where k is a hyper-parameter.

3.2.3 Interaction Site Between a Drug and a Protein

Similar to [13], a neural attention mechanism can be used between the drug molecule and the amino acid sequence which can help us understand where in the protein sequence are more likely to interact with drug molecules [36]. For the protein information the output by the multi-layer encoder, $P^{(t)} = \{p_1^{(t)}, p_2^{(t)}, \dots, p_{|P|}^{(t)}\}$, $p_i^{(t)} \in R^d$, we can calculate the importance of the drug molecule for different positions of the protein by dot-product-based scalar values.

$$h_{drug} = \delta(W_{attention}y_{drug} + b_{attention}) \quad (17)$$

$$h_i = \delta\left(W_{attention}\delta\left(W_{trans}p_i^{(t)} + b_{trans}\right) + b_{attention}\right) \quad (18)$$

$$\partial_i = tanh\left(h_{drug}^T h_i\right), \quad (19)$$

where $W_{trans} \in R^{F \times d}$, $W_{attention} \in R^{d_{atten} \times F}$, $b_{attention}$, b_{trans} are transformation parameters, δ represents the non-linear activation function. By calculating the sum of the weights of the attention mechanisms, we get the characteristic representation of the protein.

$$y_{protein} = \sum_{i=1}^{|P|} \partial_i h_i. \quad (20)$$

The characteristic representation of proteins obtained by the self-attention structure can acquire the main features of the protein according to different small molecules and enhance the robustness of the model.

3.3 Classifier and Training

Similar to previous works [13], [14], [37], [38], [39] which predict drug-target interactions based on data-driven deep learning methods, we can also concatenate representations

of drugs and proteins extracted by GAT and encoder for downstream drug-target interactions prediction. After concatenating the feature vectors extracted from small molecules and proteins, we can transform them through the fully connected layer and *softmax* function to get the possibility of interaction between them.

$$Z = W_{output} [y_{drug}; y_{protein}] + b_{output} \quad (21)$$

$$p_i = \frac{\exp(z_i)}{\sum_k z_k} i = \{0, 1\}, \quad (22)$$

where $Z \in R^2$, $W_{output} \in R^{2 \times (2 \cdot F)}$ is the weight matrix.

Because the final prediction is a supervised binary labeling problem, a cross-entropy loss function is used. In order to avoid overfitting, a regularization term is added to the expected loss function.

$$\ell(\theta) = - \sum_{k=1}^N \log p_{i_k} + \frac{\lambda}{2} \|\theta\|_2^2, \quad (23)$$

where θ is the set of all parameters in the network, N is the total number of the training samples, i_k is the k th label and λ is the coefficient of the regularization term.

4 EXPERIMENTS

In this study, we have proposed a deep learning method (called MHSADTI) for predicting DTIs. In the experiment, the AUC, Precision, Recall, AUPR and F1-score are used as the main metrics for the performance.

We perform the grid search to learn the model's best hyper-parameters. In the experiments, we initialize the weight parameters of the model using Xavier [40] which is a method of initializing weights in neural network. We optimize the training loss using the Adam [41] optimizer. The hyper-parameters of the neural networks are set as follows [13], the regularization: 1e-7, 1e-6, and 1e-5; the radius r : 0 (i.e., each atom and chemical bond), 1 and 2; the n -gram: 1, 2 and 3; the learning rate: 1e-5, 1e-4 and 1e-3; the number of layers of GAT: 2, 3 and 4; the number of layers of encoder: 1, 2; the output dimension F : 5, 10 and 20; the embedding dimension d : 10, 20 and 30. Table 2 summarizes the default parameter settings for this architecture learned by grid search.

We use Pytorch(GPU) along with Python 3.6.7 to perform our experiments. All experiments are executed on a Linux machine with processor Intel(R) Xeon(R) Gold 6230 CPU @ 2.10 GHz, 256 GB RAM, and 8 GeForce RTX 2080Ti GPU. We conduct the multiple cross validation to evaluate the prediction ability of the proposed algorithm.

4.1 Baseline

In order to assess the performance of MHSADTI, we have compared it with other two state-of-the-art methods: DeepCPI [13] and DeepConv-DTI [37], which are also used to predict DTIs. For the basic method, DeepCPI constructs a 4-layer graph convolutional network and a 4-layer convolutional network to extract characteristics of drug molecules and amino acid sequences of proteins, respectively. When obtaining feature vectors of proteins, the weighting effects of drug molecules on protein fragments are considered, and

TABLE 2
Default Parameter Settings of MHSADTI Learned by the Grid Search

Parameters	Value
Activation Function(FC)	ReLU
Activation Function(Output)	Sigmoid
Epoch	200
Optimizer	Adam
The number of layers of GAT	4
The number of encoder	1
The output dimension F	10
The embedding dimension d	10
The Regularization λ	1e-5
The learning rate	1e-3
The radius r	2
The n -gram	3
The number of multi-head self-attention	8
The decay interval	10
The learning rate decay ratio	0.5

the feature vectors of the weighted sum of protein sequences is obtained. Finally, it completes the training process within performing multi-layer perceptron structure processing on the concatenated two-part feature vectors. This method has achieved good performances in various binary classification predictions for DTIs and so it has attracted the attention of many researchers.

On the other hand, we also compare the newly proposed prediction method called DeepConv-DTI [37]. It captures local residue patterns of proteins by performing convolutional neural network on various lengths of amino acid subsequences. For the characteristic representation of drugs, it uses the Morgan/Circular drug fingerprint and each drug is represented as a 2048-dimensional binary vector, whose index indicates the existence of specific substructures. Finally, the model fully concatenates the feature vectors of drugs and proteins extracted by neural network to get prediction results after a fully connected layer.

4.2 Result

4.2.1 Effectiveness

In order to learn a generalized model, we get the experimental results with the cross validation for ten times on four datasets and take the average as the final result.

In this section, Table 3 shows the experimental results of the compared methods and MHSADTI. Similarly, we also evaluate the two MHSADTI-ablation models (MHSADTI (only gat), MHSADTI (only encoder)) shown in Table 4. As shown in Table 3, on the human dataset, MHSADTI exceeds the baseline in all evaluation metrics. Compared to DeepCPI, AUC, Precision and AUPR all achieve great improvement, from 0.9692 to 0.9822, 0.9187 to 0.9472, 0.9399 to 0.9568, respectively. Compared to DeepConv-DTI, the AUC and AUPR are 0.8 1.3 percent higher, respectively.

As shown in Table 3, on the *C.elegans* dataset, AUPR has the most obvious change, which increases from 0.9571 to 0.9832, which indicates an improvement of over the competing method. AUC, F1-score are also better than DeepCPI, from 0.9758 to 0.9838, 0.9394 to 0.9763, respectively. Compared to DeepConv-DTI, MHSADTI is also better than it in all indicators.

TABLE 3
The Results on all the Dataset: AUC, Precision, Recall, AUPR and F1-Score of the Compared Methods

Datasets	Methods	AUC (std)	Precision (std)	Recall (std)	AUPR (std)	F1-score (std)
human	RWR	0.8375 (0.008)	0.7707 (0.008)	0.7243 (0.017)	0.8165 (0.006)	0.7466 (0.010)
	DrugE-Rank	0.8562 (0.008)	0.7181 (0.015)	0.8668 (0.016)	0.8257 (0.005)	0.7851 (0.005)
	DeepConv-DTI	0.9738 (0.001)	0.9295 (0.006)	0.9175 (0.007)	0.9437 (0.009)	0.9204 (0.013)
	DeepCPI	0.9692 (0.003)	0.9187 (0.010)	0.9210 (0.008)	0.9399 (0.008)	0.9096 (0.009)
	MHSADTI	0.9822 (0.001)	0.9472 (0.007)	0.9365 (0.011)	0.9568 (0.009)	0.9346 (0.009)
<i>C.elegans</i>	RWR	0.8493 (0.007)	0.7860 (0.008)	0.7128 (0.010)	0.8212 (0.004)	0.7475 (0.005)
	DrugE-Rank	0.8221 (0.011)	0.7906 (0.011)	0.7474 (0.007)	0.8322 (0.006)	0.7684 (0.013)
	DeepConv-DTI	0.9782 (0.001)	0.9435 (0.004)	0.9423 (0.003)	0.9711 (0.004)	0.9579 (0.006)
	DeepCPI	0.9758 (0.003)	0.9393 (0.006)	0.9271 (0.009)	0.9571 (0.011)	0.9394 (0.011)
	MHSADTI	0.9838 (0.001)	0.9465 (0.009)	0.9451 (0.005)	0.9832 (0.004)	0.9763 (0.009)
DUD-E	DeepConv-DTI	0.9204 (0.006)	0.2131 (0.005)	0.6301 (0.013)	0.4264 (0.019)	0.3155 (0.025)
	DeepCPI	0.9334 (0.005)	0.2684 (0.006)	0.6425 (0.017)	0.4351 (0.016)	0.3401 (0.024)
	MHSADTI	0.9484 (0.001)	0.2462 (0.005)	0.7124 (0.009)	0.4462 (0.010)	0.3451 (0.025)
DrugBank	RWR	0.7595 (0.004)	0.7406 (0.003)	0.6511 (0.004)	0.7854 (0.002)	0.6929 (0.003)
	DrugE-Rank	0.7591 (0.004)	0.7070 (0.005)	0.6289 (0.013)	0.7598 (0.005)	0.6656 (0.009)
	DeepConv-DTI	0.8531 (0.003)	0.7891 (0.009)	0.7385 (0.014)	0.8238 (0.008)	0.7643 (0.016)
	DeepCPI	0.7003 (0.006)	0.7006 (0.003)	0.5563 (0.011)	0.7421 (0.016)	0.6552 (0.029)
	MHSADTI	0.8628 (0.012)	0.7706 (0.009)	0.7918 (0.012)	0.8351 (0.007)	0.7836 (0.011)

On the DUD-E dataset, MHSADTI achieves the greater performances than DeepCPI and DeepConv-DTI in terms of AUC. MHSADTI achieves a 1.6, 3.0 percent improvement over the DeepCPI and DeepConv-DTI, respectively.

On the DrugBank dataset, MHSADTI obtains the better results in terms of all indicators. Especially, compared to DeepConv-DTI, MHSADTI achieves a 1.1, 1.3 percent improvement in terms of AUC and AUPR, respectively.

As shown in Table 4, for the four datasets, MHSADTI gets slightly better performances than MHSADTI (only encoder) in terms of AUC. According to the AUPR, MHSADTI achieves the great improvement compared with MHSADTI (only encoder) and the improvement is more obvious when the dataset is larger. This indicates that GAT used in MHSADTI is important for improving the generalization ability of the model when the two parts of the structure are merged and processed for downstream tasks. As shown in Table 4, compared with MHSADTI(only encoder), MHSADTI has an average 0.4, 1.7 percent improvement in indicators (AUC, AUPR) on all datasets. Especially on *C.elegans* and DUD-E, MHSADTI achieves the greater improvement in terms of AUPR (from 0.9625 to 0.9832, 0.4337 to

0.4462). The results show that GAT used in MHSADTI can further enhance the robustness of the model and improve the prediction ability.

There is no doubt that extracting effective contextual information of proteins and drugs is very important for predicting DTIs. Based on the convolution, both DeepConv-DTI and DeepCPI perform well for predicting DTIs. DeepConv-DTI achieves (0.9738, 0.9782, 0.9204, 0.8531), (0.9437, 0.9711, 0.4264, 0.8238) and (0.9204, 0.9579, 0.3155, 0.7643) in terms of AUC, AUPR and F1-score for four datasets, respectively. DeepCPI achieves (0.9692, 0.9758, 0.9334, 0.7003), (0.9399, 0.9571, 0.4351, 0.7421) and (0.9096, 0.9394, 0.3401, 0.6552) in terms of AUC, AUPR and F1-score for four datasets, respectively. However, the spatial structure of proteins is complicated and there are many interactions and mutual influences between protein structures [42], [43]. The convolutional deep learning-based method is limited by the size of the neuron's perception window. Therefore, it only learns local feature information which is not sufficient. In our methods, the encoder based on multi-head self-attention not only has the ability to extract local feature information which is similar to CNN structure, but also explores the long-dependent information that may exist between different positions in the amino acid sequence. When extracting the characteristics of drugs, the possible noise connections of drug molecules are omitted by GAT based on attention mechanism. Simultaneously, the more effective and generalized characteristic information of drugs is extracted with GAT. Therefore, MHSADTI based on multi-head self-attention mechanism and GAT further improves the prediction performances and achieves AUC values of 0.9822, 0.9838, 0.9484, 0.8628 on four datasets. Furthermore, it also achieves the improvement in terms of AUPR(0.9568, 0.9832, 0.4462, 0.8351) and F1-score (0.9346, 0.9763, 0.3451, 0.7836) compared with others.

The DTI prediction is a very popular research topic and many non-deep learning related methods have been proposed in recent years. In this section, we compare two recently published methods (called DrugE-Rank [6], RWR [8]) for predicting DTIs on human, *C.elegans* and DrugBank datasets. For DUD-E, it is mainly used to evaluate methods based on molecular docking(independent test dataset). Therefore, we do not conduct experiment in DUD-E dataset about RWR and DrugE-Rank. DrugE-Rank performs an ensemble learning which integrates the prediction by diverse cutting-edge techniques and adopts different prediction methods which includes the k-nearest neighbor (kNN), BLM [2], LapRLS [44], WNN-GIP [4], [45]. RWR [8] generates the global network topology information by using random walk with restart algorithm, then the global topology information is used to train the model to predict DTIs. As shown in Table 3, MHSADTI achieves the better performances than these two methods. It also indicates that data-driven end-to-end deep learning models are more robust than non-deep learning methods because they can learn effective semantic features based on massive data.

4.2.2 Theoretical Analysis of Multi-Head Self-Attention

From the experimental results in Section 4.2.1 (shown by Table 3), we can clearly see that the model evaluation index has been improved after using the encoder structure of the

TABLE 4

The Results on all the Dataset: AUC, Precision, Recall, AUPR and F1-Score of the Two MHSADTI-Ablation Models and MHSADTI

Datasets	Methods	AUC (std)	Precision (std)	Recall (std)	AUPR (std)	F1-score (std)
human	MHSADTI	0.9716	0.9255	0.9196	0.9402	0.9120
	(only gat)	(0.003)	(0.010)	(0.008)	(0.008)	(0.010)
	MHSADTI	0.9809	0.9359	0.9419	0.9511	0.9360
	(only encoder)	(0.001)	(0.009)	(0.009)	(0.006)	(0.007)
<i>C.elegans</i>	MHSADTI	0.9822	0.9472	0.9365	0.9568	0.9346
	(only gat)	(0.001)	(0.007)	(0.011)	(0.009)	(0.009)
	MHSADTI	0.9767	0.9387	0.9293	0.9524	0.9337
	(only gat)	(0.003)	(0.009)	(0.010)	(0.008)	(0.010)
DUD-E	MHSADTI	0.9833	0.9502	0.9423	0.9625	0.9479
	(only encoder)	(0.002)	(0.006)	(0.007)	(0.008)	(0.008)
	MHSADTI	0.9838	0.9465	0.9451	0.9832	0.9763
	(only gat)	(0.001)	(0.009)	(0.005)	(0.004)	(0.009)
DrugBank	MHSADTI	0.9330	0.2110	0.6676	0.4326	0.3383
	(only gat)	(0.002)	(0.017)	(0.018)	(0.010)	(0.012)
	MHSADTI	0.9429	0.2625	0.6614	0.4337	0.3427
	(only encoder)	(0.006)	(0.007)	(0.013)	(0.004)	(0.021)
DrugBank	MHSADTI	0.9484	0.2462	0.7124	0.4462	0.3451
	(only gat)	(0.001)	(0.005)	(0.009)	(0.010)	(0.025)
	MHSADTI	0.6964	0.7087	0.5502	0.7383	0.6328
	(only gat)	(0.005)	(0.005)	(0.015)	(0.011)	(0.013)
DrugBank	MHSADTI	0.8566	0.7719	0.7832	0.8267	0.7722
	(only encoder)	(0.013)	(0.014)	(0.011)	(0.006)	(0.025)
	MHSADTI	0.8628	0.7706	0.7918	0.8351	0.7836
	(only gat)	(0.012)	(0.009)	(0.012)	(0.007)	(0.011)

multi-head self-attention mechanism. In this section, we give a theoretical analysis. As depicted in Fig. 3, it shows the information transfer process from the previous layer to the next layer in the network. Each black dot represents a vector of amino acid fragments. Different colored arrows indicate different weights for the next layer to obtain information from different positions on the previous layer. In extracting the amino acid sequence information of proteins, the CNN structure is used in the baseline and a multi-head self-attention mechanism is used in MHSADTI. First of all, compared with LSTM, both CNN and MHSADTI have the common advantage that they can perform parallel data processing. From Fig. 3, when using the CNN structure, limited by the size of the convolution window, each position in the sequence can only fuse local features within the current position range, and these features do not contain the contextual information of the sequence (the window size in Fig. 3 is 3). In this study, the encoder structure based on multi-head self-attention is used. The self-attention obtains the weight and output of this layer through the similarity

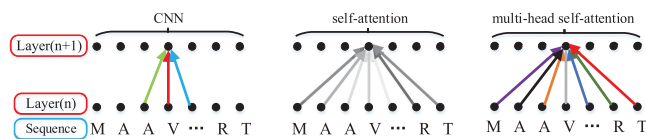


Fig. 3. Information extraction diagram of CNN, self-attention, multi-head self-attention.

TABLE 5

The Complexity, Degree of Parallelization, Maximum Step Size on Self-Attention, Multi-Head Self-Attention, Convolutional, Recurrent and the Corresponding Methods

Network Structure (each layer)	Methods	Complexity	Degree of Parallelization	Maximum Step Size
self-attention	–	$O(n^2 \cdot d)$	$O(n)$	$O(1)$
multi-head self-attention	MHSADTI	$O(n^2 \cdot d \cdot m)$	$O(n)$	$O(1)$
convolutional	DeepConv-DTI, DeepCPI	$O(k \cdot n \cdot d^2)$	$O(n)$	$O(\log_k n)$
recurrent	–	$O(n \cdot d^2)$	$O(1)$	$O(n)$

weights of the current position and different positions in sequences, avoiding the lack of learning local information of different positions in the sequence in CNN structure. After adopting the multi-head method, MHSADTI can obtain multiple layers of semantic information between different positions in the sequence, improving the robustness and stability of the model. With the “Position Encoding”, the encoder can effectively extract the long-dependent information from amino acid sequences which can not be obtained by CNN or RNN.

4.2.3 Complexity Analysis of MHSADTI

It is undeniable that the introduction of the attention mechanism can make the model more complicated, especially self-attention. In this section, we analyze the complexity of MHSADTI through two aspects: qualitative analysis [17] and quantitative analysis.

(A) *Qualitative Analysis.* As shown in Table 5, we have analyzed the computational complexity of the network structure based on self-attention, multi-head self-attention, convolution and recurrent. Different from MHSADTI, both DeepCPI and DeepConv-DTI are based on convolution. So MHSADTI takes more time than DeepCPI, DeepConv-DTI because of its attention mechanism. For an amino acid sequence of length n , it can be vectorized into an embedding matrix $A \in R^{n \times d}$ as input to models. self-attention includes three operation steps (similarity calculation, softmax and weighted average) [17]. Given $A \in R^{n \times d}$, we can get similarity matrix $S_{mi} \in R^{n \times n}$ whose elements represent the similarity between different positions. The complexity of similarity calculation is $(n, d) * (d, n) = O(n^2 \cdot d)$. The weighted average is represented as the product of the similarity matrix and the embedding matrix and its complexity is $(n, n) * (n, d) = O(n^2 \cdot d)$. As a result, the total complexity of self-attention is $O(n^2 \cdot d)$ and the complexity of multi-head self-attention that contains m self-attention is $O(n^2 \cdot d \cdot m)$. For a convolutional network with a local perception window size of k , each convolution operation is a matrix multiplication of size (n, d) and (d, k) , its complexity is $(n, d) * (d, k) = O(k \cdot n \cdot d^2)$. Therefore, the computational complexity of the RNN structure at each time step is $(n, d) * (d, 1) = O(n \cdot d^2)$.

(B) *Quantitative Analysis.* In order to have a clear understanding of the running time of the model, we have quantitatively counted the average running time of each model

TABLE 6
The Average Time Consumption (in seconds) on Extracting the Feature Vector of Drugs and Proteins, Running the Entire Model for Each Epoch

Model	Drug				Protein				All			
	human	<i>C.elegans</i>	DUD-E	DrugBank	human	<i>C.elegans</i>	DUD-E	DrugBank	human	<i>C.elegans</i>	DUD-E	DrugBank
DeepConv-DTI	2.52(0.02)	3.06(0.06)	15.4(0.64)	25.0(0.71)	7.72(0.07)	9.40(0.20)	47.4(1.97)	59.0(2.65)	42.59(0.47)	52.07(1.59)	260.8(13.1)	410.5(13.8)
DeepCPI	5.37(0.09)	4.82(0.07)	17.1(2.31)	29.6(0.59)	6.58(0.11)	6.99(0.11)	24.1(2.80)	41.1(0.90)	258.6(0.65)	266.1(0.42)	735.7(11.6)	1241(4.65)
MHSADTI	10.7(1.09)	10.8(0.54)	51.3(0.98)	59.5(0.63)	11.9(0.79)	13.0(0.68)	59.6(1.04)	68.7(0.81)	111.3(6.97)	117.3(4.73)	567.7(7.63)	615.1(6.91)

during 200 epochs. The running time of each epoch represents the time consumption required for the model to complete the training of all data once. In the experiment, we ensure that all controllable environmental factors (early stop mechanism, parallelization degree and so on) are the same. As shown in Table 6, each column corresponding to “Drug” and “Protein” represents the time (in seconds) taken for the forward propagation of the modules in the model to extract the feature vectors of drugs and proteins. We can see from Table 6 that MHSADTI takes more time due to the use of GAT based on the attention mechanism in the “Drug” column. Similarly, MHSADTI also takes more time due to multi-head self-attention for extracting feature vectors of proteins in “Protein” column. In summary, in the process of

forward propagation, the feature extraction of drugs based on GAT and the feature extraction of proteins based on the encoder take more time and it is consistent with the qualitative analysis of Table 5. The “All” column represents the sum time of forward propagation and backward propagation. The backward propagation contains a complicated process of continuous derivative function. Table 6 indicates that MHSADTI can take less time than DeepCPI in an average epoch. It seems inconsistent with the theoretical analysis in Table 5, and the reason may be that the backward propagation of the deep learning-based method exists unpredictability. Considering the improvement of prediction performance, it should be acceptable for MHSADTI to take more time than DeepConv-DTI.

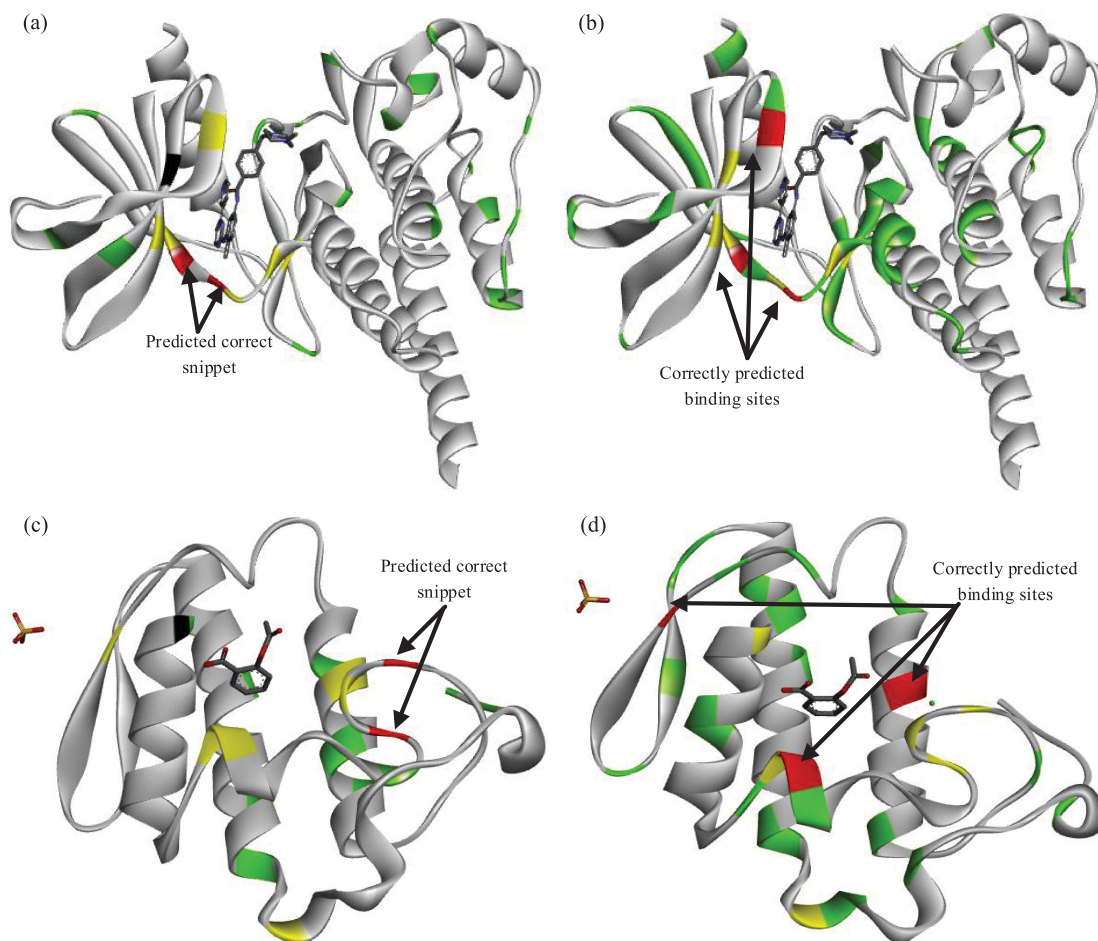


Fig. 4. The complex of imatinib and Tyrosine-protein kinase SYK (PDB ID:1XBB) [(a), (b)], aspirin and Phospholipase A2 (PDB ID: 1TGM)[(c), (d)]. (a) and (c) are visualizations of DTI with attention weights of multi-head self-attention, the regions in proteins with high weight values are highlighted in red and they are predicted correctly. (b) and (d) are visualizations of DTI with attention weights of drug-protein attention, the regions in protein with high weight values are highlighted in red and they are correctly predicted as binding sites.

4.2.4 Case Study for Interpretability

In this study, while giving effective improvement results, the interpretability of the learning content of the two self-attention models is visualized. To demonstrate this, those regions with high weight values calculated from the two attention models are mapped onto a known 3D protein structure. Fig. 4 shows the complex of imatinib and Tyrosine protein kinase SYK (PDB ID: 1XBB) and the complex of aspirin and Phospholipase A2 (PDB ID: 1TGM).

(a) and (c) shown in Fig. 4 are the visualization results of the high-weight parts learned by the multi-head self-attention mechanism. The black fragment is a known binding site. As explained by Equations (12), we use the representation vectors of the query and key-value pairs in the encoder to calculate the weight of the remaining positions of the amino acid sequence to its feature vector. The yellow fragments are the known binding sites and should have a high weight for the representation of the black fragment. The red fragments are the high-weight areas and the known binding positions that we correctly predicted through the calculated weight value. The green fragments are predicted locations that have a high weight for black fragments but are not known binding sites. Through this way of calculating the self-attention, it can be found which fragments have a strong connection with each other, and the strong or weak connections between such fragments can effectively help the discovery of drugs.

(b) and (d) shown in Fig. 4 are the visualization results of the high-weight areas learned by the drug-protein attention mechanism. In the 3D structure of proteins, the part that is not a binding site while predicted to be a binding site is highlighted in green. The parts marked as yellow are true binding sites while not giving a high weight score. The red part is the position with a high weight score predicted by the algorithm and the real binding site for drug action. It can be seen that the algorithm can effectively capture the binding sites.

5 CONCLUSION

Accurately predicting DTIs can greatly improve efficiency and reduce cost during the drug development. Although advanced studies have greatly improved the accuracy of DTI predictions, numerous challenges remain in the prediction of DTIs.

In this study, we have proposed an end-to-end neural network model (called MHSADTI) for predicting DTIs. MHSADTI extracts characteristics of drug molecules and proteins by GATs and encoder based on multi-head self-attention. We have assessed the performance of the proposed method by conducting multiple cross validation. The computational experimental results show that MHSADTI can obtain better performance than the baseline. More importantly, the visualization of weight scores learned by the multi-head self-attention in the method confirms the effectiveness of this learning method.

Although MHSADTI has obtained valid experimental results, it is clear that it still has shortcomings for being further improved. First, in the data input to the model, we only use the one-dimensional data representation in the structural characteristic information of drugs and proteins (SMILES and amino acid sequences). However, drugs and proteins actually

have very complicated spatial structures. Although one-dimensional data representation is easy to obtain, it does lose the high-level characteristic information of many drugs and proteins in prediction. Second, in the input of the model, we only use a biological representation data for drugs and proteins. It is believed that integrating more comprehensive biological data to the deep learning model and eliminating various noises are very helpful for more accurate predictions of DTIs. We would consider integrating more useful data to deep learning models in the future.

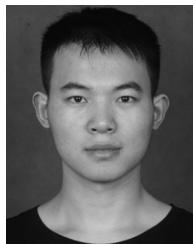
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